

ORACLE CONSULTANCY · CHIEF-OF-STAFF SYSTEM

Company System Master Guide

The big picture — how the whole system is conceived, structured and run. The architectural & governance companion to the day-to-day Operations Manual.

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- Philosophy · architecture · data model · roles · governance · roadmap
 - V3 · 18 June 2026 · For the owner, directors and any future operator/maintainer

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This guide stays at altitude — the step-by-step "how do I..." lives in the **Operations Manual**; the at-a-glance card is the **Quick Reference**; the feature inventory is **What We've Built**.

01 The idea

Why the system exists

A single Chief-of-Staff command centre that runs the day-to-day operations of Oracle Consultancy and its seven portfolio companies — for one operator, replacing a sprawling Excel workbook with a living system that has a memory.

The brief was not "digitise the spreadsheet" but "give the operator a brain": capture work and never lose its history, keep people and compliance current, remember every meeting, and surface the whole portfolio on one glanceable brief — so decisions are made from one source of truth.

Code	Company	Code	Company
CO01	Dar Spices	CO05	PES Ltd
CO02	Cocozuri Chocolat	CO06	MES Ltd
CO03	Terra Green	CO07	Pamoja Plus
CO04	Oracle Consultancy (also the parent brand)		

02 Design principles

The rules everything follows

Principle	What it means in practice
Reuse, don't duplicate	One capability underpins many features: Documents power compliance, tasks/todos power checklists, OECR underpins assets, Outbox powers messaging, Home/Brief surface signals. New features extend existing tables rather than inventing parallel ones.
One design language — "Aurora"	Every page, dialog and form shares the same liquid-glass surfaces, single blue accent, centred layout, iPhone-style toggles and calm motion. Learn one screen, know them all.
Single source of truth	Append-only history and audit trails; facts are never overwritten (the fact ledger keeps versions). Access can be revoked but data is never deleted.
Glanceable, every-number-a-door	Counts and figures are clickable; the system shows where to look, then lets you act.
Portal parity	The staff portal is first-class — it shares visuals and motion with admin, dropping only what would expose other people's data.
Grounded intelligence	AI works from your records and cites them; it never invents data, and degrades gracefully when switched off.

03 How it all fits together

Architecture

A single Next.js application, server-rendered, talking to one Postgres database, hosted close to that database, with a thin AI layer bolted on the side.

The stack

Layer	Technology
App	Next.js 16 (App Router), React 19, TypeScript 5 — one codebase serves admin + portal
Data	Supabase Postgres via the pooler (port 6543); Drizzle ORM + postgres.js; <code>supabase-js</code> for newer write paths
Styling	Tailwind v4 tokens (<code>globals.css</code>) + the Aurora component kit
AI	Groq Cloud (<code>llama-3.1-8b-instant</code>) for language; pgvector + an embedding Edge Function for semantic search
Hosting	Vercel, functions pinned to the EU region to sit beside the database (kills transatlantic latency)

Request flow



Every request passes the edge gate in `src/proxy.ts` (the Next-16 `proxy` convention), which checks the signed session cookie before any admin or portal page renders. Reads and writes happen in server components and server actions — there is no separate API tier to keep in sync.

The AI layer

AI is additive, not load-bearing. A safety harness (`src/lib/ai-json.ts`) wraps every Groq call with retries, timeouts and JSON guards; a master switch turns it all off cleanly. Semantic search indexes tasks, documents, meetings and people into pgvector and blends vector hits with full-text search.

CONFIGURATION THAT MUST NOT CHANGE

`prepare:false` + `max:1` in `src/db/index.ts` (PgBouncer transaction mode); the pooler port 6543; the `secret()` derivation must stay identical across `proxy.ts`, `admin-auth.ts` and `portal-auth.ts`; all wall-clock columns are `timestampz` written as UTC and shown in local (Dar es Salaam, UTC+3) time.

04 The data model at a glance

What the database holds

Everything hangs off a small set of reference tables (companies, departments, sites, roles) and the people/tasks core. Full detail is in [memory/database_schema.md](#).

Area	Key tables
Core	companies · departments · people · tasks (+ task_assignees, task_updates)
Reference	sites · job_titles · department_heads · reporting_lines · person_companies
Meetings	meetings · meeting_tasks
Governance (transfer-pack)	facts (ledger) · cap_table · beneficial_owners · key_persons · signatories · resolutions · risks · decisions · pipeline · commitments
HR compliance	requirement_profiles · requirement_items · person_requirements
HR ops	assets · asset_assignments · vendors · leave_types · leave_requests · attendance · public_holidays · todos
Documents	documents · document_links · inbox
Letters / stock / cleaning	letters · stock_items/purchases/issues · cleaning_areas/days/checks
Comms	chat_threads · chat_participants · chat_messages · chat_message_mentions · reminders · outbox · notifications
Auth & system	webauthn_credentials · settings · audit_log · corrections · system_events · daily_snapshots · undo_tokens

SCHEMA DISCIPLINE

Schema changes go through Drizzle migrations (edit `schema.ts` → generate → review → `npm run db:migrate`), always after a backup. The live DB has some historical drift, so migrations use `IF NOT EXISTS` where needed.

05 Roles & the access model

Who can do what

Two separate worlds, two separate cookies: the owner's admin side (`cos_admin`) and the staff portal (`cos_portal`).

Role	Scope	Can do
Owner	Everything, all companies	The full admin command centre. One password (+ optional owner identity 2nd factor) or a passkey.
Staff	Own data only	Own tasks, documents, attendance, leave, profile.
Manager	Own + their team / company	Team attendance, leave approvals, team roster, create tasks for their company.
HR	All companies (tasks)	Group-wide task visibility & creation.
Director	All companies (operator)	Board brief; create tasks/events; draft messages. Auto-routed to the board.

DATA-OWNERSHIP RULE

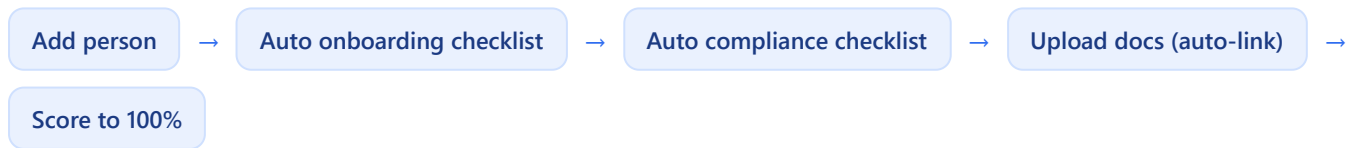
Revoking access, changing a role or archiving a person never deletes their data — access control and records are independent. This keeps history and compliance intact.

Sign-in security: signed HttpOnly cookies verified at the edge; a session "generation" lets the owner invalidate all other devices by changing the password; passkeys (WebAuthn) store only public keys.

06 Core workflows end-to-end

How the pieces connect

A new staff member joins



A meeting becomes action



A document keeps compliance current



Someone leaves



A letter goes out



Each workflow reuses shared building blocks, so a change to (say) the compliance engine improves the person drawer, the staff portal and the Director Brief at once.

07 Operating rhythm & governance

Running it well

Cadence	Focus
Daily	Work tasks, clear notifications, approve leave, action needs-attention rows, glance at Home health.
Weekly	Review the Director Brief; chase compliance gaps; check pipeline + commitments for notice-due items.
Monthly	Review attendance and completed/closed work; confirm backups are healthy.
Before any DB change	<code>npm run db:backup</code> , then migrate.

Governance controls

- **Director outreach pause/resume** — a kill-switch in Settings for director messaging.
- **Audit everywhere** — task history, person history, compliance events, `system_events`.
- **Transfer pack** — board-level governance data (cap table, owners, resolutions, risks) kept current for handover.

08 Reliability, security & maintenance

Keeping it safe

Backups	Portable JSON snapshots (<code>npm run db:backup</code> / <code>db:restore</code>) + Supabase cloud backups (primary).
Monitoring	Sentry (errors only) when the DSN is set.
Performance	56 DB indexes; EU-region functions; cached reference getters.
Quality	42+ Vitest safety tests on money/leave/compliance/status maths; <code>tsc</code> type-check (4 GB heap).
Dependencies	Dependabot + pinned <code>overrides</code> (clean <code>npm audit</code>).

TWO HARD "DON'TS"

Never `rm -rf .next` while the dev server runs (corrupts the build cache). Never change the cookie `secret()` derivation in one place without the others — it locks sessions out.

09 Roadmap & pending decisions

What's next

The system is feature-complete for daily operations. The open items are mostly **owner decisions** that unlock already-built capability, plus a few planned enhancements.

Owner actions that switch things on

Item	What's needed
Email automation	Add Gmail/Resend credentials in Vercel (sending is already wired).
WhatsApp automation	Twilio business verification + number + approved templates → set 3 env vars (sandbox already proven).
Semantic search	Run the one-time setup (<code>SEMANTIC_SEARCH.md</code>) and flip the Settings toggle.
Announcements noticeboard	Review the built A1–A3 feature and approve shipping.
Dependabot alerts / Supabase Pitr	Enable in the respective dashboards.

Planned (designed, not built)

- Durable multilingual semantic search (Swahili/Hindi/Gujarati) via a small EU container — a cost decision (~£3–7/mo).
- Director send-as-self email; if-this-then-that outreach rules.
- Further premium "operating brain" phases (see `memory/premium_upgrade_plan.md`).

WHERE THE TRUTH LIVES

This guide is a snapshot. The authoritative, continuously-updated record is `CLAUDE.md` and the `memory/` folder (plans, audits, per-area deep dives), with the design rules in `DESIGN_SYSTEM.md` .